Business Problem Statement & Project Overview

1. Understanding the Business Problem

Product availability is essential to converting customer interest into sales. For an online fashion retailer such as ASOS, unavailable products and sizes may result in missed purchases and reduced customer satisfaction.

This project examines product availability across the ASOS catalog to identify stockout patterns, compare brands, and highlight inventory areas that warrant further investigation.

2. Identifying Inventory and Supply Chain Challenges

The analysis focuses on differences in size availability across products and brands. It investigates which brands have the highest stockout rates, how these rates vary with product prices, and where unavailable sizes may represent significant sales opportunities.

Stockouts are treated as indicators of potential inventory constraints. Establishing their causes requires additional information, such as sales history, replenishment schedules, and supplier lead times.

3. Purpose of the Data Analysis

The project aims to turn product-level pricing and availability data into practical insights for inventory planning and merchandising.

The key objectives are to:

- Measure stockout counts and rates across products and brands.
- Compare brand-level pricing and availability patterns.
- Develop a consistent proxy for potential revenue exposure.
- Identify priorities for replenishment reviews and purchasing decisions.

4. Data Preparation and Cleaning Using Python

The initial stage involves preparing a dataset containing more than 18,000 product records from products.csv, using Python and Pandas in Google Colab.

Data preparation includes resolving parsing issues, converting prices into numeric values, handling missing data, and standardizing relevant fields. Brands with very few product records are filtered from comparative analysis to reduce the influence of small samples.

5. Feature Engineering and Revenue Exposure Modelling

Text-processing techniques transform unstructured product information into features suitable for analysis. These include standardized brand names, counts of unavailable sizes, and product-level stockout rates.

A price-weighted stockout metric is calculated as:

Revenue Exposure Proxy = Product Price × Number of Unavailable Sizes

This metric helps prioritize products and brands for review. It does not measure actual lost revenue, because the dataset does not establish customer demand, missed purchases, or the number of units that would otherwise have sold.

6. Data Visualization and Brand Quadrant Analysis

Matplotlib and Seaborn are used to visualize inventory patterns and compare brand performance.

A **Brand Strategy Quadrant** plots average product price against stockout rate, with bubble sizes representing the revenue exposure proxy. This highlights brands with different combinations of price and availability, helping stakeholders identify where further investigation or replenishment review may be useful.

7. Business Reporting and Recommendations

The findings are presented in a structured inventory assessment covering stockout patterns, brand comparisons, and potential revenue exposure.

Recommendations focus on reviewing replenishment priorities, investigating persistent availability gaps, and validating purchasing decisions against sales and margin data. Decisions about expanding brand investment or liquidating

inventory require additional evidence on demand, profitability, inventory age, and turnover.

8. Project Documentation and GitHub Repository

The project deliverables—including Python notebooks, data preparation steps, feature definitions, visualizations, and a README—are organized in a structured GitHub repository.

Documentation explains the analytical workflow, assumptions, metric limitations, and instructions for reproducing the results, supporting clear review by stakeholders and portfolio reviewers.